Asymmetric Volatility, Regime Dependence, and Tail-Risk Calibration in MSCI U.S. Equity Factors

Evidence from GARCH/EGARCH, VaR backtests, and conditional Mean-Variance allocation

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Roadmap

1. Motivation & contribution: why volatility modeling is the bottleneck in factor risk
2. Data & stylized facts: non-normality, high correlations, and crisis co-movement
3. Volatility modeling: GARCH vs EGARCH, leverage effects, persistence
4. Regime dependence: conditional efficiency across volatility states
5. Tail-risk calibration: 95% vs 99% VaR (Kupiec) and what fails in the extreme tail
6. Portfolio construction: conditional covariance $\Rightarrow$ frontier compression and "diversification collapse"
7. COVID-19 as a stress laboratory: linking asymmetry, correlation convergence, VaR rejection, and frontier compression
8. Practical implications & research extensions

Motivation & contribution

Core question

When markets enter extreme volatility states, do standard symmetric volatility models (GARCH) mis-calibrate factor risk in a way that matters for allocation and VaR governance?

- The paper uses MSCI U.S. equity factor indices (Value, Momentum, Quality, MinVol, Size) and SPX as the market proxy.
- Empirical focus: (i) 2014–2026 "full sample" regime behavior; (ii) COVID-19 2019–2021 as an extreme-volatility case study.
- Output focus: volatility asymmetry (EGARCH), regime-conditioned efficiency, VaR backtesting, and conditional mean–variance allocation.

Contributions

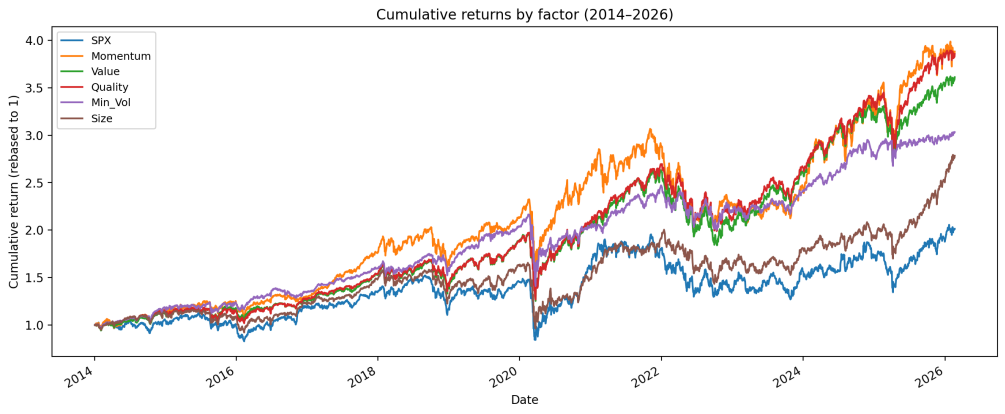
1. **Unified empirical narrative** linking four crisis mechanisms measured on the same factor universe:
   - volatility asymmetry (EGARCH leverage),
   - volatility persistence (half-life),
   - correlation convergence in stress (dependence tightening),
   - and **portfolio-level frontier compression** under conditional covariance.
2. **Governance-relevant diagnostics**: VaR passes at 95% for most series but *systematically fails at 99%* (Kupiec), concentrating model risk where oversight is most stringent.
3. **Actionable translation**: model-implied conditional covariance produces an investable allocation map (Equal Weight vs Vol Parity vs MinVar) that reveals when "diversification" becomes structurally constrained.

Data & stylized facts

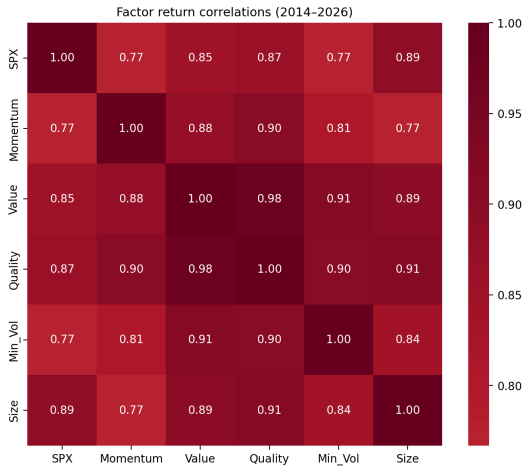
Data, universe, and design

- Universe: SPX (market) + five MSCI U.S. factor indices: Momentum, Value, Quality, Minimum Volatility, Size.
- Frequency: daily returns. Horizon: 2014–2026 for the long-sample analysis; 2019–2021 for the COVID stress window.
- Modeling stack:
  - univariate volatility: GARCH(1,1) and EGARCH(1,1),
  - regime split by volatility state (Low/Med/High),
  - tail calibration: dynamic VaR and Kupiec unconditional coverage,
  - portfolio layer: conditional mean–variance cloud (5,000 long-only portfolios) under model-implied covariance.

Stylized fact #1: Factor premia exist, but crisis co-movement is the dominant story



Stylized fact #2: Unconditional correlations are high (diversification is limited ex ante)



Volatility modeling: GARCH vs EGARCH

Modeling intuition

Symmetric GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

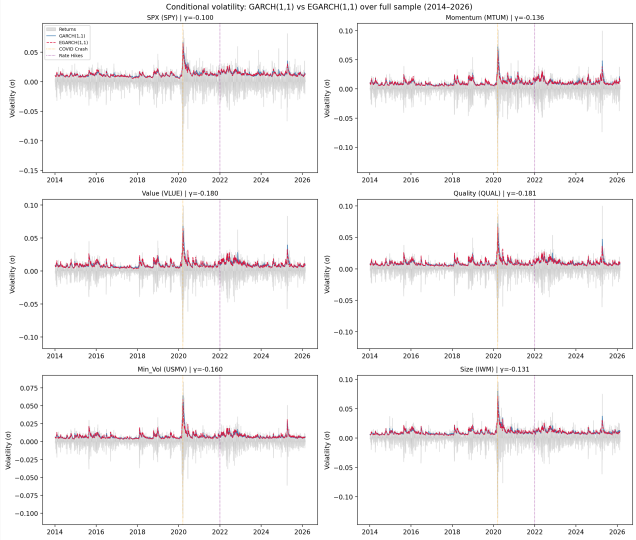
- Captures volatility clustering.
- Imposes *symmetry*: $+\varepsilon$ and $-\varepsilon$ of same size have identical impact.

Asymmetric EGARCH(1,1)

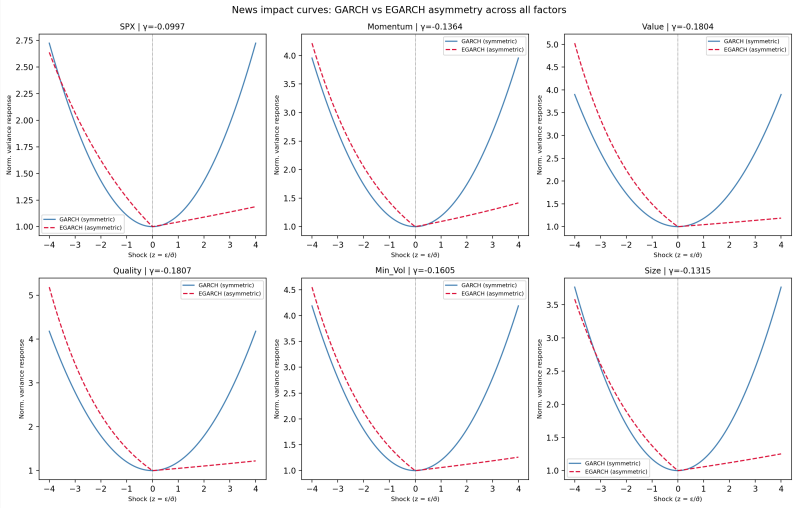
$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha\left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - \mathbb{E}|z|\right) + \gamma\frac{\varepsilon_{t-1}}{\sigma_{t-1}}$$

- $\gamma < 0$: negative shocks increase volatility *more* than positive shocks.
- Directly targets leverage/asymmetry documented in the empirical volatility literature.

Conditional volatility paths: GARCH vs EGARCH (2014–2026)



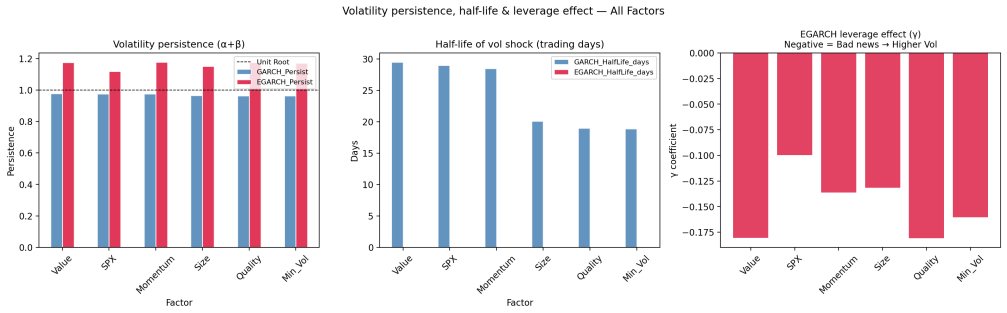
Asymmetry is systematic: News impact curves (GARCH vs EGARCH)



Persistence, half-life, and leverage (cross-factor summary)

Key empirical message: GARCH persistence is high (slow mean reversion), while EGARCH leverage is negative across all series.

- GARCH half-lives are measured in *weeks*, not days.
- Leverage ranks are economically meaningful: Value/Quality show the largest negative γ in the project estimates.



Volatility regimes: design and conditional efficiency

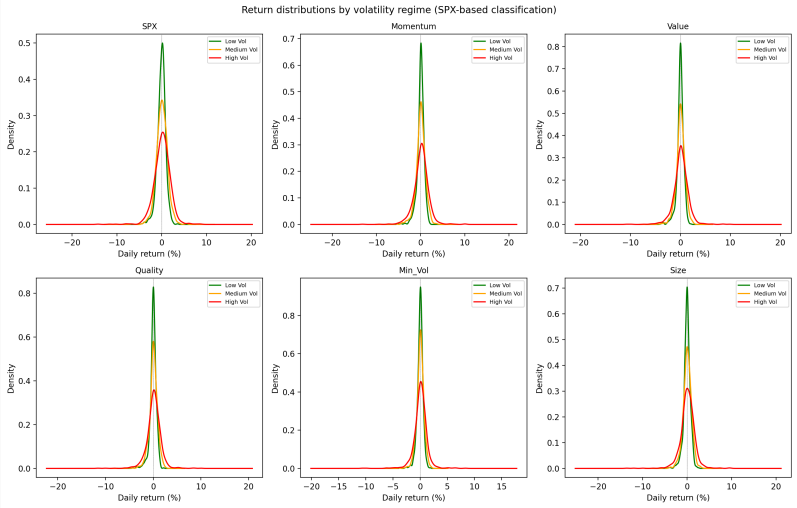
Regime design and why it matters

- **State variable.** Daily volatility regime is defined using the *GARCH(1,1) conditional volatility* of the market sleeve (SPX proxy), $\hat{\sigma}_t^{\mathsf{SPX}}$.
- **Tercile classification.** Each day t is assigned to:

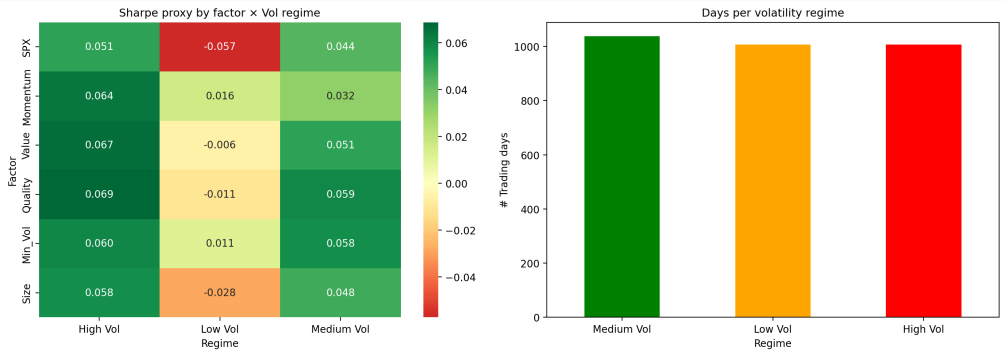
$$\begin{aligned} \text{Low Vol} &: \hat{\sigma}_t^{\mathsf{SPX}} \leq Q_{33} \\ \text{Medium Vol} &: Q_{33} < \hat{\sigma}_t^{\mathsf{SPX}} \leq Q_{67} \\ \text{High Vol} &: \hat{\sigma}_t^{\mathsf{SPX}} > Q_{67} \end{aligned}$$

  where Q_{33} and Q_{67} are the 33rd and 67th percentiles of $\hat{\sigma}_t^{\mathsf{SPX}}$ over the full sample.
- **Within-regime diagnostics.** For each factor and regime, we compute mean return, volatility, a Sharpe proxy (mean/vol), and regime-specific max drawdown.
- **Interpretation.** This design converts volatility persistence into economically meaningful "states" and provides an allocation-relevant lens: conditional efficiency and diversification are evaluated *where the portfolio actually lives* (in regimes that persist for weeks, not days).

Return distributions shift materially across regimes



Conditional Sharpe proxy by regime (heatmap) + time spent in each regime

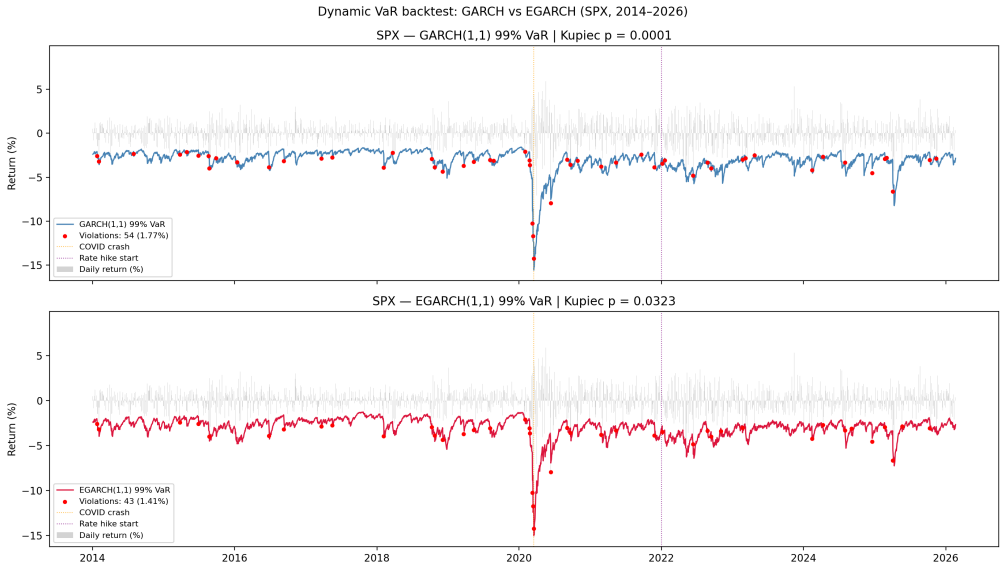


Tail risk calibration: VaR backtests

Why VaR backtesting is the governance bottleneck

- Passing VaR at 95% is not informative about robustness at 99%.
- The project's Kupiec tests show that mis-calibration concentrates in the *extreme* left tail.
- This is consistent with the paper's descriptive evidence: fat tails + leverage + crisis clustering.

Dynamic 99% VaR backtest (SPX): GARCH vs EGARCH



Kupiec unconditional coverage: 95% vs 99% (summary)

Panel A: 95% VaR

- Breach rates cluster near 5% for most factors.
- Momentum is the systematic exception (higher breaches and lower p-values).

Panel B: 99% VaR

- Breach rates are *significantly above* 1% across the universe under both models.
- EGARCH improves SPX coverage (higher p-value) but does not fully eliminate rejection.

Kupiec VaR backtests

Panel A: 95% VaR

Factor	G br.(%)	EG br.(%)	G p	EG p
SPX	4.72	5.01	0.47	0.97
Momentum	5.80	5.83	0.05	0.04
Value	4.72	5.11	0.47	0.78
Quality	5.41	5.54	0.31	0.18
Min_Vol	4.92	4.95	0.83	0.90
Size	5.01	4.92	0.97	0.83

Panel B: 99% VaR

Factor	G br.(%)	EG br.(%)	G p	EG p
SPX	1.77	1.41	0.00	0.03
Momentum	2.10	1.97	0.00	0.00
Value	2.03	2.00	0.00	0.00
Quality	2.10	1.90	0.00	0.00
Min_Vol	1.61	1.70	0.00	0.00
Size	1.97	1.90	0.00	0.00

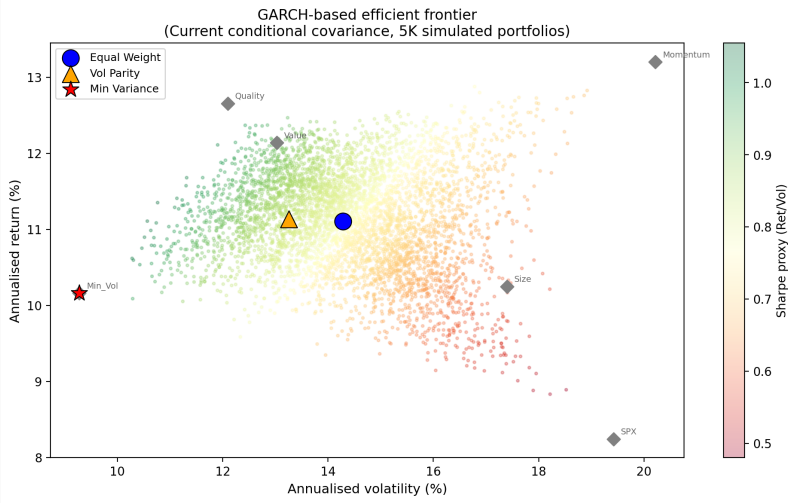
Footnote (method): Kupiec (1995) POF test compares observed exceedances to the binomial expectation under correct unconditional coverage; under the null, the LR statistic is asymptotically $\chi^2(1)$.

Portfolio construction: conditional frontier compression

From volatility modeling to allocation: why conditional covariance matters

- The relevant covariance matrix for allocation is *state-dependent* (volatility persistence + correlation convergence).
- Under elevated conditional correlations, the optimizer's feasible set compresses: the incremental variance reduction from diversification becomes small.
- This is observable as a **frontier compression** in the conditional mean–variance opportunity set.

GARCH-based efficient frontier (current conditional covariance)



Model-implied portfolio weights (long-only)

Factor	Cond. Vol (% ann.)	Equal (%)	Vol Parity (%)	MinVar (%)
SPX	19.42	16.67	12.09	0.00
Momentum	20.22	16.67	11.62	0.00
Value	13.02	16.67	18.04	0.00
Quality	12.10	16.67	19.42	0.00
Min_Vol	9.27	16.67	25.33	100.00
Size	17.40	16.67	13.50	0.00

Covariance construction. The portfolio covariance used for the long-only minimum-variance solution is $\Sigma_t = D_t \bar{R} D_t$, where $D_t = \mathrm{diag}(\hat{\sigma}_{t,1}, \ldots, \hat{\sigma}_{t,N})$ contains **current** GARCH conditional volatilities (annualized) and $\bar{R}$ is the **long-run** correlation matrix estimated over the full sample. Volatility parity uses weights proportional to $1/\hat{\sigma}_{t,i}$. No train/test split is applied in the allocation exercise: the figure is a point-in-time allocation implied by the full-sample estimates.

Interpretation: “diversification collapse” as a mechanical outcome

1. **High correlations** imply limited diversification potential across style sleeves.
2. **Heteroskedasticity & persistence** keep relative volatility differences in place long enough to matter for allocation.
3. **Result:** long-only MinVar concentrates in Min_Vol (lowest conditional variance), revealing a practical limit of mean–variance diversification inside a single equity factor complex.

Link back to tail risk: in the same states where correlations tighten and volatility spikes, 99% VaR rejects more strongly (frontier compression and VaR failure are two expressions of the same structural regime shift).

COVID-19 as an extreme volatility case study

COVID-19 narrative: why this episode is a clean stress test

- COVID concentrates extreme volatility, forced deleveraging, and correlation tightening into a short window.
- It provides a high-signal setting to evaluate whether symmetry assumptions (GARCH) induce economically meaningful underestimation of risk.
- Empirical lens: (i) residual non-normality; (ii) leverage; (iii) peak-volatility underprediction; (iv) backtest rejection.

Connecting the four mechanisms

1. **Asymmetry:** EGARCH news impact curves show stronger variance response to negative shocks across all factors.
2. **Persistence:** shocks decay slowly (weeks) $\Rightarrow$ volatility states persist.
3. **Dependence:** correlations are already high; in stress they tighten further (documented in the paper's regime analysis).
4. **Portfolio & risk systems:** conditional frontier compresses, while 99% VaR fails coverage more systematically than 95%.

Takeaway: risk is not only higher in crise, it becomes *more one-dimensional* (common shock dominates), which undermines both diversification and tail calibration.

Implications & conclusion

Implications for PMs, risk managers, and academics

For portfolio managers

- Treat factor premia as *state-contingent* exposures: conditional efficiency differs across volatility regimes.
- Expect diversification benefits to shrink when they are most valuable; plan hedges and liquidity buffers accordingly.

For risk managers

- A 95% VaR "pass" is not a comfort metric; 99% is where model risk concentrates.
- EGARCH improves calibration but extreme-tail misspecification persists (fat tails + clustered shocks).

For academics

- The evidence motivates heavy-tailed innovations, regime-switching volatility, and multivariate asymmetric covariance models to jointly address asymmetry and dependence tightening.

Conclusion

- MSCI U.S. equity factors exhibit **systematic leverage effects** and **high volatility persistence**.
- Dependence is structurally elevated, limiting unconditional diversification and enabling **frontier compression** under conditional covariance.
- VaR backtests show a sharp governance result: **models are broadly acceptable at 95% but fail at 99% across the factor complex**.
- EGARCH improves risk calibration relative to symmetric GARCH (notably for SPX), but **extreme-tail risk remains under-modeled**, implying the need for heavier tails and/or explicit regime/tail overlays.

Bottom line: Factor diversification is a fair-weather concept unless risk systems explicitly model asymmetry, persistence, and crisis dependence in an integrated framework.

Appendix

Kupiec (1995) POF statistic (reference form)

$$LR_{\mathrm{POF}} = -2\ln\left(\frac{(1-\alpha)^{N-x}\,\alpha^{x}}{(1-\hat{p})^{N-x}\,\hat{p}^{x}}\right), \qquad \hat{p} = \frac{x}{N}$$

- N: number of forecasts; x: number of VaR exceedances; α: nominal tail probability.
- Under correct unconditional coverage, $LR_{\mathrm{POF}} \overset{a}{\sim} \chi^2(1)$.